

LCA Overview of comparing Multiple-use Paper Grocery Bags to Multiple-Use Plastic Grocery Bags

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Project Goals



What is the purpose of the study?

- The purpose of this study is to compare the Life Cycle Assessment of multiple-use paper bags and multiple-use plastic grocery bags.
- Calculate the impacts of paper and plastic bag and evaluate which bag has the lowest impact over its lifespan

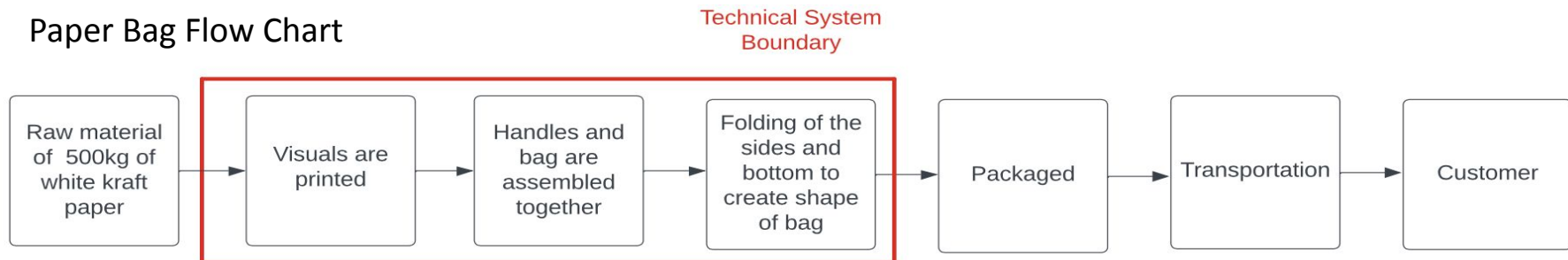
LCA Study Scope & Flow Chart

-Functional unit: 1 kg of groceries

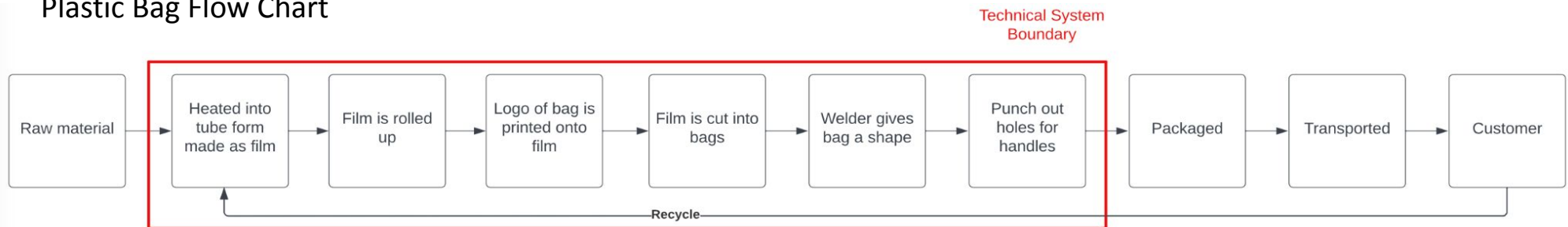
-Impact Categories: Global Warming and Land Use

- LCA Type: Comparative

Paper Bag Flow Chart



Plastic Bag Flow Chart

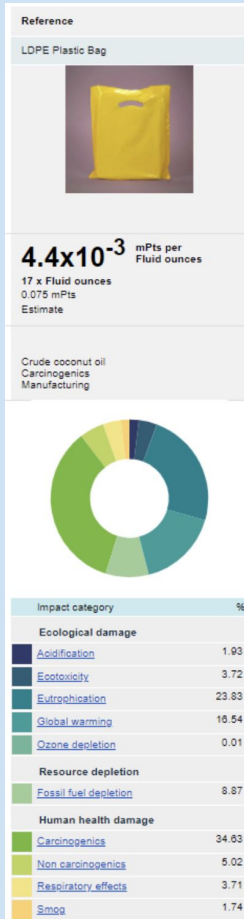


Eco-Design Strategies

- Low impact materials (Renewable Resources and avoid materials that damage ecological health)
- Optimized end-of-life (Ability to be recycled and/or biodegrade)



LDPE Plastic Bag



- LDPE stands for low density polyethylene
- Most common form of grocery bags
- Can be recycled into many different plastic products, like trash can liners and shipping envelopes.

Wow!

System BOM of Plastic Bag

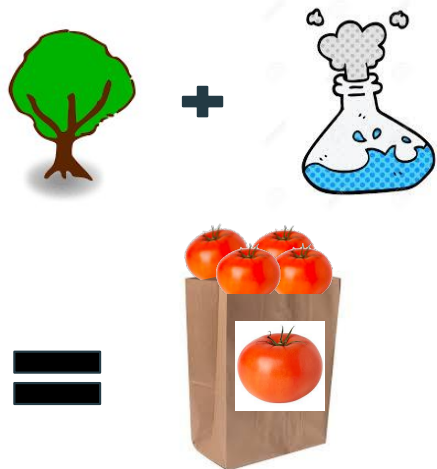
Name	Material/Process	Qty	Amt	Unit	mPts	CO ₂ eq. kg	MS	Part ID
 Part <i>Printing colour for a plastic bag.</i>	Printing colour, rotogravure	1	0.035	g	1.04x10 ⁻⁵	1.06x10 ⁻⁴	E	
 Copy of Part <i>Crude oil is used as a raw material</i>	Crude coconut oil	1	0.9	kg	0.0503	0.239	E	
—  Polyethylene LDPE <i>Polyethylene LDPE is a part of a c</i>		1	47	g	0.0154	0.272	E	
 Material	Polyethylene, LDPE, granu	47		g	7.21x10 ⁻³	0.0982	E	
 Process	Extrusion, plastics	47		g	1.31x10 ⁻³	0.0238	E	
 Process	Stretch blow molding, plast	47		g	4.39x10 ⁻³	0.100	E	
 Process	Thermoforming, plastics	47		g	2.50x10 ⁻³	0.0497	E	
Manufacturing total					0.0657	0.511	E	

Name	Transportation mode	Qty	Amt	Unit	mPts	CO ₂ eq. kg	MS	Part ID
— Assembled product								
 Transportation mode <i>Used a truck that transports plast</i>	Truck, 20-28t	119		mi	2.72x10 ⁻³	0.0375	E	
— Sub-assemblies and parts								
 Part <i>Printing colour for a plastic bag.</i>	Printing colour, rotogravure, 5	1	0.035	g			E	
 Copy of Part <i>Crude oil is used as a raw materi</i>	Crude coconut oil	1	0.9	kg			E	
 Polyethylene LDPE <i>Polyethylene LDPE is a part of a</i>	Polyethylene, LDPE, granulat	1	47	g			E	
Transportation total					2.72x10⁻³	0.0375	E	

Name	End of life method	Qty	Amt	Unit	mPts	CO ₂ eq. kg	MS	Part ID
 Part <i>Printing colour for a plastic bag.</i>	Printing colour, rotogravure, 5	1	0.035	g			E	
 Copy of Part <i>Crude oil is used as a raw material ex</i>	Crude coconut oil	1	0.9	kg			E	
—  Polyethylene LDPE <i>Polyethylene LDPE is a part of a cher</i>		1	47	g	0	0	E	
 Material	Polyethylene, LDPE, granulat	47		g			E	
 Process	Recycling	47		g	0	0	E	
End of Life total					--	--	E	

Name	Consumables/water/power	Amt	Unit	mPts	CO ₂ eq. kg	MS
—  Consumables				0.167	11.6	E
 Consumable <i>Total amount of CO2 emitted from a j</i>	Carbon dioxide, emitted during use	0.00026	t	3.40x10 ⁻³	0.236	E
—  Water use				7.60x10 ⁻⁷	4.25x10 ⁻⁶	E
 Water <i>Water used for a plastic bag.</i>	Water, completely softened	0.0387	gal	7.60x10 ⁻⁷	4.25x10 ⁻⁶	E
—  Power use				2.91x10 ⁻³	0.0741	E
 Power <i>Energy used for a plastic bag.</i>	Heat from propane/butane	0.509	MJ	2.91x10 ⁻³	0.0741	E
Use total				0.170	11.6	E

Paper Bag



- Kraft paper bags are one of the most commonly used paper bags for groceries.
- Typically composed of 30-100% recycled content
- Intended for single use but can be reused
- Kraft process is Chemical pulping which produces higher quality and stronger paper.



System BOM of Paper Bag

Name	Consumables/water/power	Amt	Unit	mPts	CO ₂ eq. kg	MS
Consumables				5.23x10 ⁻⁶	3.63x10 ⁻⁴	E
Consumable	Carbon dioxide, emitted during use	0.0008	t	5.23x10 ⁻⁶	3.63x10 ⁻⁴	E
Water use				1.97x10 ⁻⁵	1.10x10 ⁻⁴	E
Water	Water, completely softened	1.004	gal	1.97x10 ⁻⁵	1.10x10 ⁻⁴	E
	1.004 gallons per paper bag.					
Power use				0.0150	0.382	E
Power	Heat from propane/butane	2.622	MJ	0.0150	0.382	E
	Total amount of energy used for a paper bag.					
Use total				0.0150	0.382	E

Name	End of life method	Qty	Amt	Unit	mPts	CO ₂ eq. kg	MS	Part ID
Printing Process	Ink, soy-based, ASA, black	1	20	g			E	
	Printing Process of a paper bag							
epichlorohydrin	Epichlorohydrin, from hypochlorite	1	0.9	g			E	
	Epichlorohydrin used as a part of							
adhesive		1	26.5	g	0.0344	1.11	E	
	Glue used for a paper bag.				0.0344			
Material	Adhesive; EVA, waterbased		26.5	g			E	
Process	Landfill, plastics, mixture		26.5	lb	0.0344	1.11	E	
urea formaldehyde		1	0.1	g	0	0	E	
	urea formaldehyde is used as a chemical							
Material	Urea formaldehyde resin		0.10000000149011612	g			E	
Process	Recycling		0.10000000149011612	g	0	0	E	
Melamine		1	0.00376	g	0	0	E	
	Melamine is a chemical used for							
Material	Melamine		0.003759999992325902	g			E	
Process	Recycling		0.003759999992325902	g	0	0	E	
one Kraft unbleached paper		1	54.4	g	0	0	E	001
	one Kraft unbleached paper bag							
Material	Kraft paper, unbleached		54.400001525878906	g			M	
Process	Recycling		54.400001525878906	g	0	0	E	
	Recycling of paper bag once							
End of Life total					0.0344	1.11	E	

Name	Transportation mode	Qty	Amt	Unit	mPts	CO ₂ eq. kg	MS	Part ID
Assembled product								
Truck	Truck, 20-28t		119	mi	2.93x10 ⁻⁴	4.04x10 ⁻³	E	
	Truck and trailer that would transport							
Sub-assemblies and parts								
one Kraft unbleached paper	Kraft paper, unbleached	1	54.4	g			E	001
	one Kraft unbleached paper bag							
Printing Process	Ink, soy-based, ASA, black	1	20	g			E	
	Printing Process of a paper bag							
epichlorohydrin	Epichlorohydrin, from hypochlorite	1	0.9	g			E	
	Epichlorohydrin used as a part of							
urea formaldehyde	Urea formaldehyde resin	1	0.1	g			E	
	urea formaldehyde is used as a chemical							
adhesive	Adhesive; EVA, waterbased	1	26.5	g			E	
	Glue used for a paper bag.							
Melamine	Melamine	1	0.00376	g			E	
	Melamine is a chemical used for							
Transportation total					2.93x10 ⁻⁴	4.04x10 ⁻³	E	

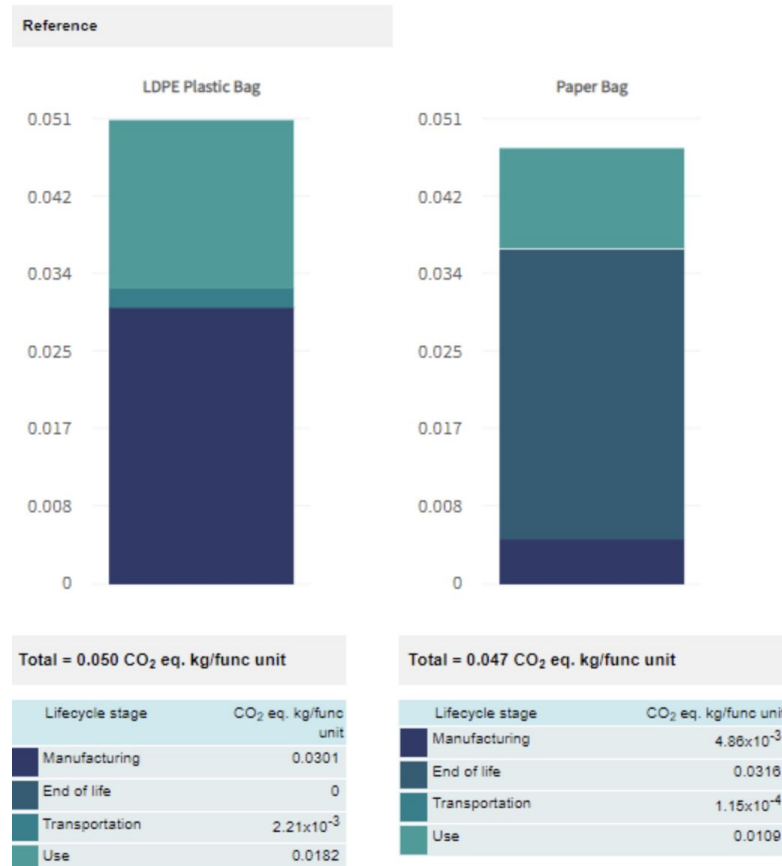
Name	Material/Process	Qty	Amt	Unit	mPts	CO ₂ eq. kg	MS	Part ID
adhesive	Adhesive; EVA, waterbased	1	26.5	g	2.77x10 ⁻³	0.0362	E	
	Glue used for a paper bag.							
urea formaldehyde	Urea formaldehyde resin	1	0.1	g	2.86x10 ⁻⁵	3.10x10 ⁻⁴	E	
	urea formaldehyde is used as a chemical							
one Kraft unbleached paper	Kraft paper, unbleached	1	54.4	g	5.03x10 ⁻³	0.0695	E	001
	one Kraft unbleached paper bag							
Printing Process	Ink, soy-based, ASA, black	1	20	g	4.18x10 ⁻³	0.0603	E	
	Printing Process of a paper bag							
epichlorohydrin	Epichlorohydrin, from hypochlorite	1	0.9	g	3.01x10 ⁻⁴	3.79x10 ⁻³	E	
	Epichlorohydrin used as a part of							
Melamine		1	0.00376	g	1.76x10 ⁻⁶	2.64x10 ⁻⁵	E	
	Melamine is a chemical used for							
Material	Melamine		0.003759999992325902	g	1.46x10 ⁻⁶	2.02x10 ⁻⁵	E	
Process	Injection molding, plastic		0.003759999992325902	g	2.94x10 ⁻⁷	6.22x10 ⁻⁶	E	
Manufacturing total					0.0123	0.170	E	

Plastic Bag vs Paper Bag

Paper bags score higher on their impact for acidification, global warming, and fossil fuel depletion potential.

Plastic bags score higher for their ecotoxicity and eutrophication potential.

Impacts by life cycle stage: Carbon footprint [CO₂ eq. kg/func unit]



Impact per Functional Unit

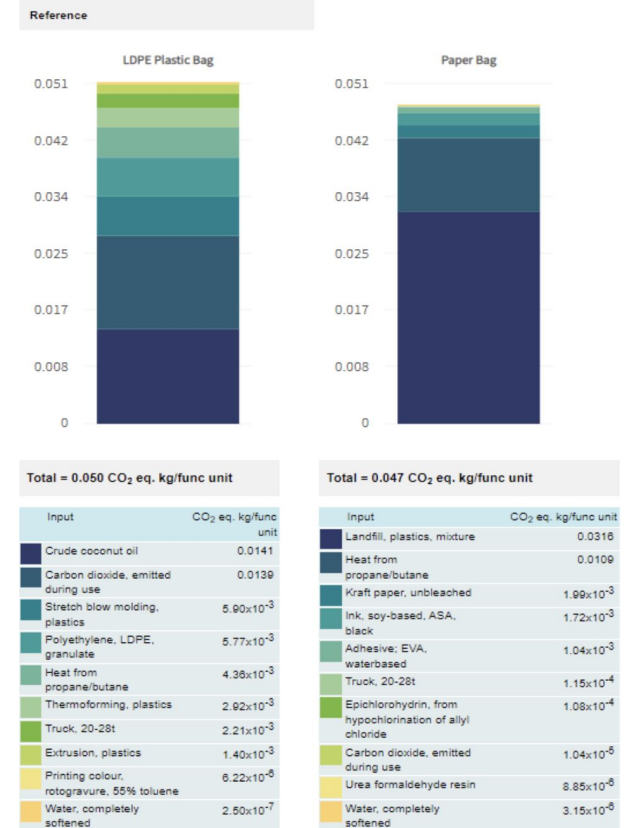
- Paper Bag: 4.4x10⁻³
- Plastic Bag: 1.8x10⁻³

Performance Improvement

- 60% based off plastic bag (reference)

Impacts by SBOM inputs: Carbon footprint [CO₂ eq. kg/func unit]

Display inputs as



Conclusion

- Although the Paper bag manufacturing process consumes more material and energy than plastic bags, the sustainable mind program concluded that it has less impact on the environment because of its recyclability and biodegradability.
- 60% Performance Improvement from LDPE plastic bag.
- Paper bag has a lower impact in terms of its functional unit.
- Paper bags should be used more widely than plastic

Functional unit: Fluid ounces		Impacts / functional unit mPts/func unit	CO ₂ eq. kg / functional unit CO ₂ eq. kg/func unit	Performance improvement from reference mPts	Performance improvement from reference %	Units of svc delivered Svc. Units	Assessment type
Create a new Concept + Reference  LDPE Plastic Bag Copy Declare as: Final		4.4x10 ⁻³	0.050			17	Estimate
Final, Lowest impact  Paper Bag Copy Delete Declare as: Reference		1.8x10 ⁻³	0.047	+2.6x10 ⁻³	+60%	35	Estimate